



BAULKHAM HILLS HIGH SCHOOL

2016

**HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION**

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In Questions 11–16, show relevant mathematical reasoning and/or calculations
- Marks may be deducted for careless or badly arranged work

Total marks – 100

Exam consists of 13 pages.

This paper consists of TWO sections.

Section I – Page 2-5 (10 marks)

- Attempt Question 1-10
- Allow about **15** minutes for this section

Section II – Pages 6-13 (90 marks)

- Attempt questions 11-16
- Allow about **2 hours and 45** minutes for this section

The reference sheet is on page 14.

Section I

10 marks

Attempt questions 1-10

Allow about 15 minutes for this section.

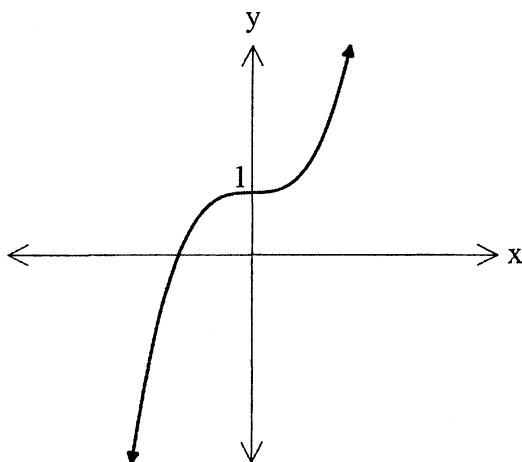
Use the multiple choice answer sheet for questions 1-10

1. What is 0.0050279 written in scientific notation correct to 3 significant figures?

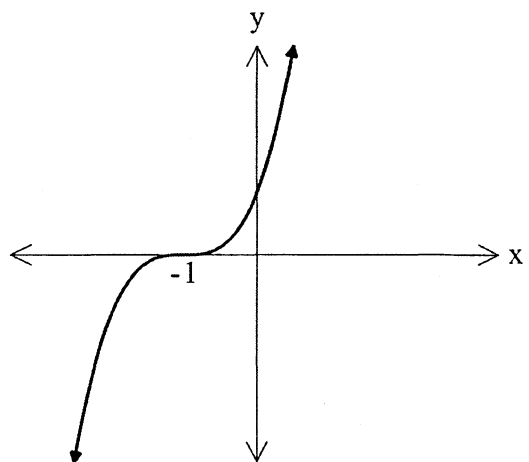
- (A) 5.02×10^{-2}
- (B) 5.03×10^{-2}
- (C) 5.02×10^{-3}
- (D) 5.03×10^{-3}

2. The graph of $y = 1 - x^3$ could be:

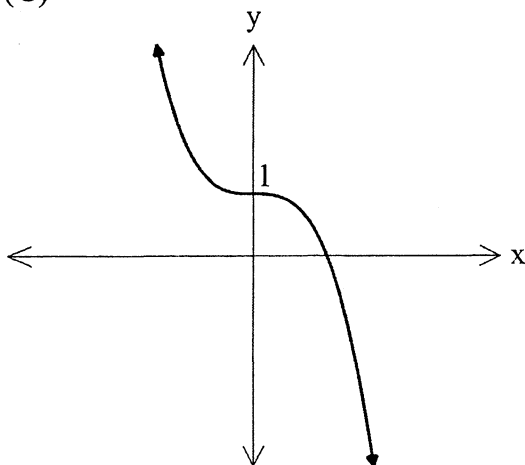
(A)



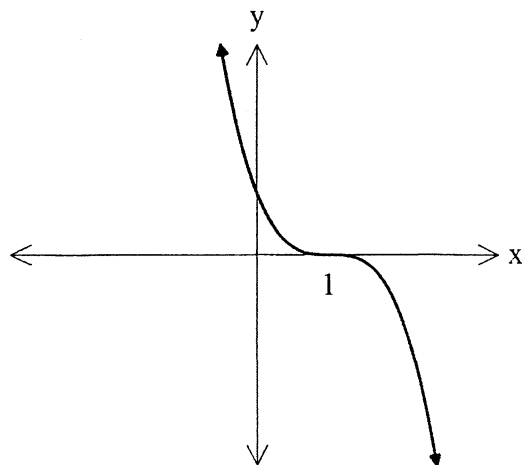
(B)



(C)



(D)



3. What is the perpendicular distance of the point $(2, -1)$ from the line $y = 3x + 1$?

(A) $\frac{6}{\sqrt{10}}$

(B) $\frac{6}{\sqrt{5}}$

(C) $\frac{8}{\sqrt{10}}$

(D) $\frac{8}{\sqrt{5}}$

4. $(2\sqrt{5} - \sqrt{3})^2 =$

(A) 17

(B) 23

(C) $17 - 4\sqrt{15}$

(D) $23 - 4\sqrt{15}$

5. A parabola has a focus of $(-3, 0)$ and a directrix of $x = 1$. What is the equation of the parabola?

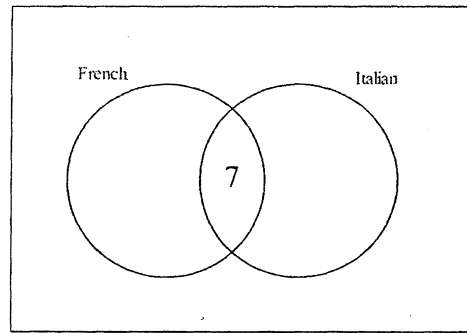
(A) $y^2 = 16(x + 3)$

(B) $y^2 = -16(x + 3)$

(C) $y^2 = 8(x + 1)$

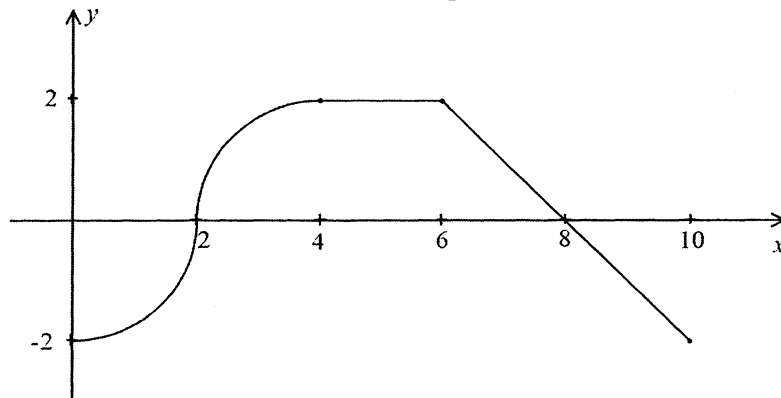
(D) $y^2 = -8(x + 1)$

6. The Venn Diagram represents a group of 30 students all of whom study either Italian, French or both languages. 7 students study both and 19 students study French.



By completing the Venn Diagram or otherwise, find the probability that if 2 people are selected at random that they only study Italian.

- (A) $\frac{11}{87}$
- (B) $\frac{2}{145}$
- (C) $\frac{22}{145}$
- (D) $\frac{57}{145}$
7. The graph of $y = f(x)$ is drawn below, showing two quadrants and two line segments.



Which of the following is true?

- (A) $\int_0^{10} f(x) dx = 2\pi + 8$ and the curve is differentiable at $x = 2$
- (B) $\int_0^{10} f(x) dx = 4$ and the curve is differentiable at $x = 2$
- (C) $\int_0^{10} f(x) dx = 2\pi + 8$ and the curve is not differentiable at $x = 2$
- (D) $\int_0^{10} f(x) dx = 4$ and the curve is not differentiable at $x = 2$

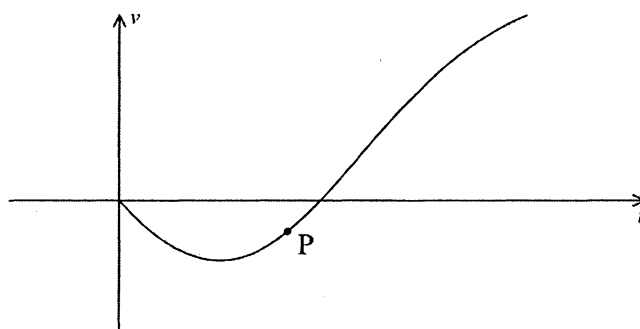
8. Which inequality defines the domain for $y = \frac{1}{\sqrt{x^2-9}}$?

- (A) $x < -3$ or $x > 3$
- (B) $x \leq -3$ or $x \geq 3$
- (C) $-3 < x < 3$
- (D) $-3 \leq x \leq 3$

9. What is the value of $\int_1^4 \frac{1}{3x} dx$?

- (A) $\frac{1}{3} \ln 3$
- (B) $\frac{1}{3} \ln 4$
- (C) $\ln 9$
- (D) $\ln 12$

10. The graph shows the velocity of a particle moving along a straight line as a function of time.



Which statement describes the motion of the particle at the point P .

- (A) The particle is moving left at increasing speed.
- (B) The particle is moving left at decreasing speed.
- (C) The particle is moving right at decreasing speed.
- (D) The particle is moving right at increasing speed.

END OF SECTION I

Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours and 45 minutes for this section

Answer each question in the appropriate page in the writing booklet.

In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

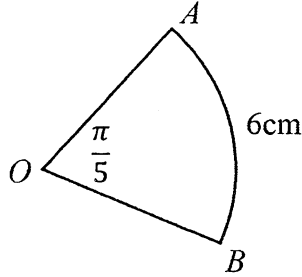
Question 11 (15 marks) - Start on the appropriate page in your answer booklet	Marks
a) Simplify	
(i) $\frac{1-x}{3} + \frac{x}{4}$	1
(ii) $\frac{8^{x-2}}{2^{3-x}}$	2
b) Factorise fully $16 - 36x^2$	2
c) Two 6 sided dice are rolled. What is the probability that a 5 or a 4 is on the upper most face?	2
d) Differentiate	
(i) $2 \sin 3x$	2
(ii) $(2x + 1)^8$	2
e) Evaluate $\int_0^2 e^{2x} dx$	2
f) Find $\int 2\cos\pi x dx$	2

End of Question 11

Question 12 (15 marks) - Start on the appropriate page in your answer booklet.

Marks

- a) The sector shown has arc length of 6 cm and $\angle AOB$ is $\frac{\pi}{5}$ radians.



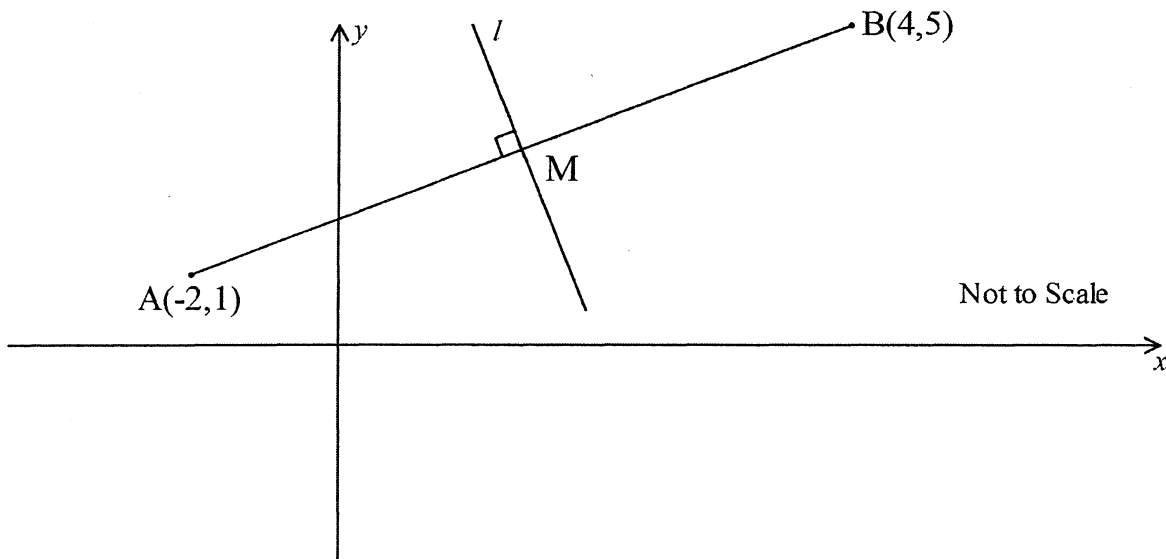
- (i) Find the radius of the circle in terms of π .

1

- (ii) Hence, find the exact area of the sector.

1

- b) The interval AB is drawn where $A = (-2,1)$ and $B = (4,5)$ and the line l is perpendicular to AB passing through M .



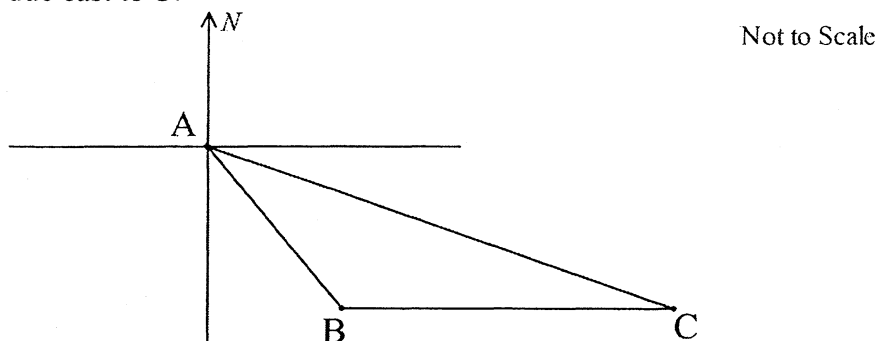
- (i) Show that M the midpoint of AB has coordinates $(1,3)$.

1

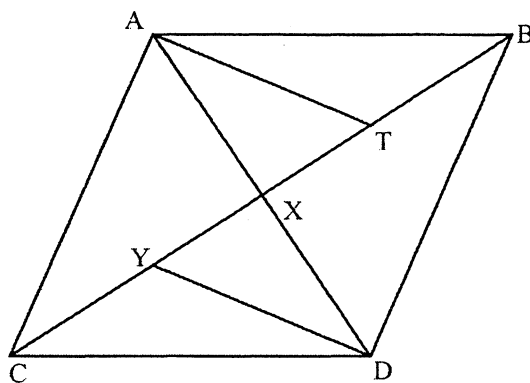
- (ii) Show that the equation of the line l , perpendicular to AB passing through M , is given by $3x + 2y - 9 = 0$.

3

- c) The diagram below, represents the journey taken by a ship which leaves point A and travels 200km on a bearing of 112° to B . It then turns and travels 150 km due east to C .



- (i) Draw a neat sketch of the diagram above in your answer booklet. 1
 - (ii) Show $\angle ABC = 158^\circ$ 2
 - (iii) Find the distance AC .
- d) The roots of the quadratic equation $x^2 - kx - 6 = 0$ are α and β . Find k if $\alpha^2\beta + \alpha\beta^2 = 4$. 2
- e) $ABDC$ is a rhombus whose diagonals intersect at X . Y and T lie on BC such that $AT \parallel DY$.



- (i) Draw a neat sketch of the diagram above in your answer booklet.
- (ii) Prove $\triangle AXT \equiv \triangle XYD$. 3
- (iii) Prove $ATDY$ is a parallelogram. 1

End of Question 12

Question 13 (15 marks) - Start on the appropriate page in your answer booklet

Marks

a) Consider the curve $y = x^3 - 6x^2 - 15x + 2$

- | | | |
|-------|---|----------|
| (i) | Find the stationary points and determine their nature. | 3 |
| (ii) | Find the point of inflexion. | 2 |
| (iii) | Sketch the curve labelling the stationary points, the point of inflexion and y intercept. | 2 |

b) 150 koalas were introduced to an isolated island.

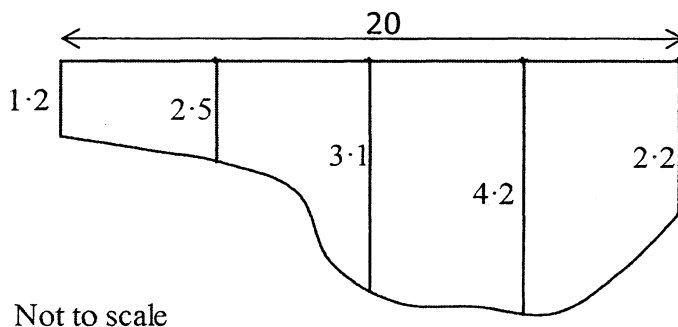
The rate at which the population (P) of the koalas increase is proportional to the population according to the differential equation,

$$\frac{dP}{dt} = 0.02P \text{ where } t \text{ is measured in years.}$$

- | | | |
|------|--|----------|
| (i) | Show $P = Ae^{0.02t}$ is a solution to the differential equation above. | 1 |
| (ii) | How many koalas are on the island after 10 years? | 1 |
| (ii) | How many years will it take for the population of the koalas to reach 200? | 2 |

c) At a certain location, a river is 20 metres wide. At this location the depth of the river in metres has been measured at 5 metre intervals.

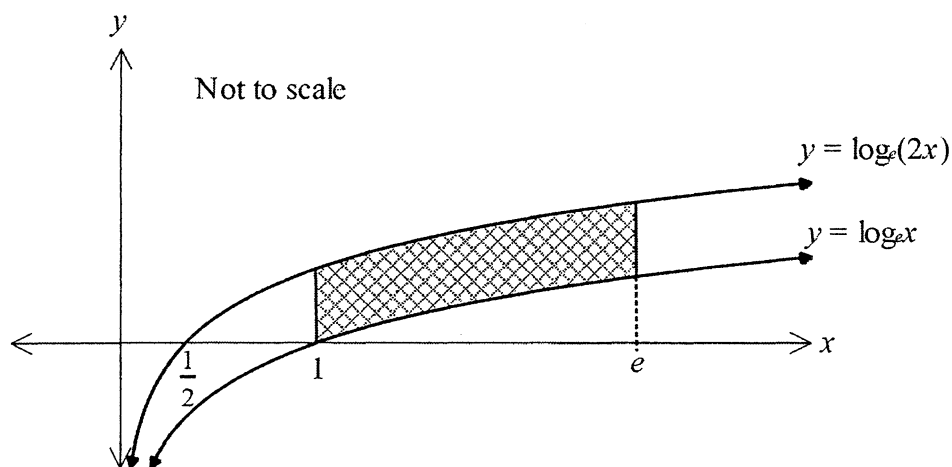
The cross section of the river is shown below.



- | | | |
|------|--|----------|
| (i) | Use Simpson's Rule with the 5 depth measurements to calculate the approximate area of the cross section. | 3 |
| (ii) | The river flows at 0.6m/sec. Calculate the approximate volume of water flowing through the cross section of the river in 10 seconds. | 1 |

End of Question 13

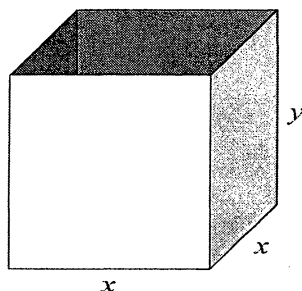
- a) The acceleration of a particle moving along the x axis is given by $\ddot{x} = 4 \sin 2t$ where x is the displacement from the origin in metres and t is the time in seconds. Initially the particle is at the origin moving to the left at 1 metre/second
- (i) Show the velocity is given by $\dot{x} = 1 - 2 \cos 2t$. 1
- (ii) Find the time when the particle first comes to rest. 2
- (iii) Show that $x = t - \sin 2t$. 1
- (iv) Find the distance travelled by the particle in the first $\frac{\pi}{2}$ seconds. 3
- b) (i) Show that $e^{1-\ln 2} = \frac{e}{2}$ 1
- (ii) Find the equation of the tangent to the curve $y = e^{1-4x}$ at $x = \frac{\ln 2}{4}$ 3
- c) The curves of $y = \ln x$ and $y = \ln 2x$ are drawn below. 4



Find the shaded area between the curves $y = \ln 2x$ and $y = \ln x$ and the lines $x = 1$ and $x = e$.

End of Question 15

- a) A rectangular box with a square base and no top is drawn below.



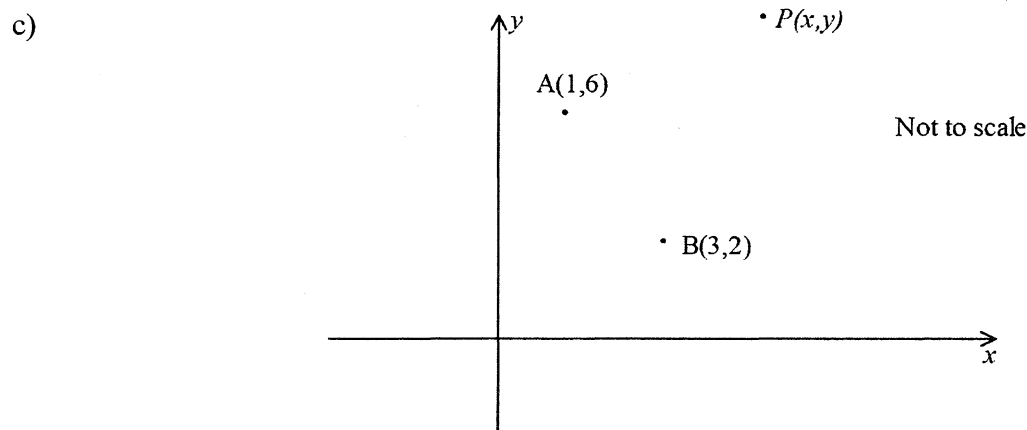
The volume of the box is 500cm^3

- (i) Show that the surface area (A) of the box is given by $A = x^2 + \frac{2000}{x}$. 2

- (ii) Find the least area of sheet metal required to make the box. 3

- b) (i) The sequence $\sin \theta, 2 \cos \theta, 2 \sin \theta \dots$ form the first three terms of an arithmetic progression.
Find the value of θ to the nearest minute. 3

- (ii) Find the next term in the sequence in terms of $\sin \theta$. 2



- (i) Find the gradient of PA in terms of x and y . 1

- (ii) The point P moves such that the gradient of PB is twice the gradient of PA . Find the values of a, b and c when the locus is expressed in the form 3

$$y = a + \frac{b}{x-c}$$

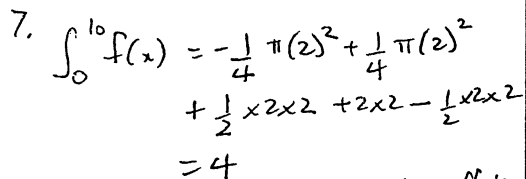
- (iii) Sketch the locus of P . 1

End of Exam

1. D 2. C 3. C 4. D 5. D 6. A
7. D 8. A 9. B 10. B

$$d = \frac{|3(2) + (-1)(-1) + 1|}{\sqrt{3^2 + (-1)^2}} = \frac{8}{\sqrt{10}} \quad (c)$$

5/  $v = (-1, 0) \ a > 2$
 $(y-0)^2 = -4(2)(x+1)$
 $y^2 = -8(x+1) \quad (c)$



Tangent is vertical $\therefore$ not diff'ble.
 $\therefore$ (D)

$$\begin{aligned} 9. \int_1^4 \frac{1}{3x} dx &= \frac{1}{3} \int_1^4 \frac{1}{x} dx \\ &= \frac{1}{3} (\ln x) \Big|_1^4 \\ &= \frac{1}{3} (\ln 4 - \ln 1) \\ &= \frac{1}{3} \ln 4 \end{aligned}$$

∴ moving left & slowing down (B)

$$\begin{aligned} \text{(ii)} \quad \frac{8^{x-2}}{2^{3-2x}} &= \frac{2^{3x-6}}{2^{3-2x}} \rightarrow \textcircled{1} \\ &= 2^{3x-6-3+2x} \\ &= 2^{4x-9} \quad \textcircled{1} \end{aligned}$$

$$= 4(2 - 5x)(2 + 5x)$$

$$P(\text{4 or 5 on upper face})$$

$$= \frac{20}{36} = \frac{5}{9} \textcircled{1}$$

(1 mark for $\left(\frac{24}{36} = \frac{2}{3}\right)$
 $\left(\frac{22}{36} = \frac{11}{18}\right)$)

d) i) $\frac{d}{dx} (2 \sin 3x) = 6 \cos 3x$

$$(ii) \frac{d}{dx} [(2x+1)^8] = 8(2x+1)^7 \times 2$$
$$= 16(2x+1)^7 \quad (1)$$

$$e) \int_0^2 e^{2x} dx = \frac{1}{2} [e^{2x}]_0^2 \quad (1)$$

$$= \frac{1}{2} (e^4 - 1) \quad (1)$$

$$f) \int 2 \cos \pi x \, dx = \frac{2}{\pi} \sin \pi x + C$$

a) ii) $l = r \therefore r = \frac{6}{\frac{\pi}{5}} = \frac{30}{\pi} \text{ ①}$
(9.549)✓

$$\begin{aligned} \text{(ii)} \quad A &= \frac{1}{2} r^2 \theta \\ &= \frac{1}{2} \cdot \left(\frac{30}{\pi} \right)^2 \cdot \frac{\pi}{5} \\ &= \frac{900\pi}{10\pi^2} = \frac{90}{\pi} \quad (28.648) \end{aligned}$$

$$b) (1) M = \left(\frac{-2+4}{2}, \frac{5+1}{2} \right) = (1, 3)$$

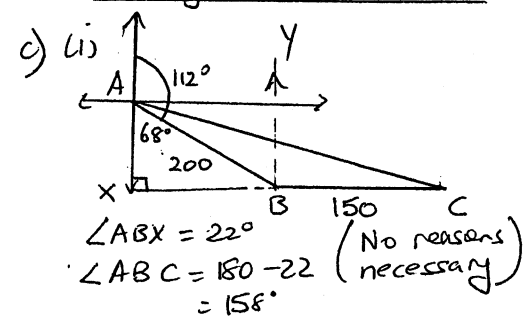
$$(ii) m_{AB} = \frac{5-1}{4-2} = \frac{2}{3}$$

$$I_m = -\frac{3}{2} \quad \text{--- (1)}$$

$$y - 3 = -\frac{3}{2}(x - 1) \quad (i)$$

$$2y - 6 = -3x + 3 \quad \text{--- (1)}$$

$$\underline{3x + 2y - 9 = 0}$$



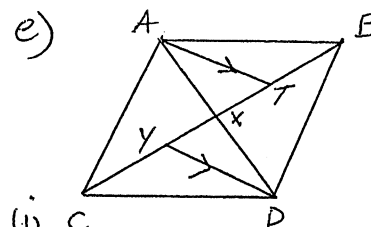
$$\therefore \angle A B Y = 68^\circ \quad \angle Y B C = 90^\circ$$

$$\therefore \angle ABC = 15^\circ$$

$$(ii) \quad AC^2 = 200^2 + 150^2 - 2(200)(150)\cos 158^\circ \quad (1)$$

$$AC = 343.7 \text{ km} \quad (1)$$

d) $x^2 - kx - 6 = 0$ $\alpha + \beta = k$
 $\alpha^2\beta + \alpha\beta^2 = 4$ $\alpha\beta = -6$ ①
 $\alpha\beta(\alpha + \beta) = 4$
 $-6k = 4$ $k = -\frac{2}{3}$ ②



$AX = X \cdot C$ (diagonals of a rhombus bisect each other) (

$$\left. \begin{aligned} \angle TAX &= \angle XDY \text{ (Alternate } \angle\text{'s)} \\ \angle XYD &= \angle ATX \text{ (" " " ")} \end{aligned} \right\} \text{ (i)}$$

$$\therefore \Delta ATX \equiv \Delta XYC \text{ (AAS) } \textcircled{1}$$

(ii) $XY = TX$ (Matching sides in Δ 's)

$\therefore AD$ and TY bisect each other

• ATOY is a parallelogram
(diagonals bisect each other)
both statements must be
made for (1).

Question 13.

a) $y = x^3 - 6x^2 - 15x + 2$

(i) st. pts $y' = 0$

$y' = 3x^2 - 12x - 15$

$3(x^2 - 4x - 5) = 0$ ①

$3(x-5)(x+1) = 0$

$x = -1, 5$

$y = 10 \quad y = -98$

$(-1, 10) \quad (5, -98)$ ①

Test nature $y' = 6x - 12$

when $x = -1 \quad y' = -18 < 0$
 $\therefore \max$ ①

when $x = 5 \quad y' = 18 > 0$
 $\therefore \min.$ ①

(ii) Point of inflexion when
when $y'' = 0$ + concavity
change.

$y'' = 6x - 12 = 0$

when $x = 2$ ①

Test change in concavity

at $x = -1 \quad y'' = -18 < 0$

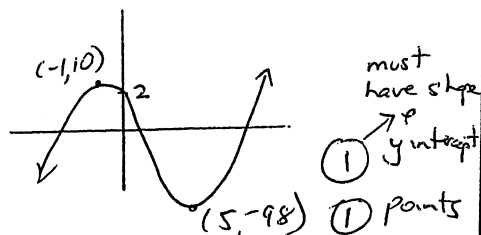
at $x = 5 \quad y'' = 18 > 0$ ①

change in concavity $\therefore$
point of inflexion.

bus $P = Ae^{0.02t}$ - ①

$\frac{dP}{dt} = 0.02(Ae^{0.02t})$

$\therefore \frac{dP}{dt} = 0.02P$



(i) See bottom of first column.

b) (i) $P = P_0 e^{0.02t}$

when $t = 0 \quad P = 150$

$\therefore 150 = P_0 e^0 \rightarrow P_0 = 150$

$P = 150 e^{0.02t}$ ①

when $t = 10$

$P = 150 e^{0.2}$ ①

$P = 183$ (ie 183 konkas)

(ii) find t when $P = 200$

$200 = 150 e^{0.02t}$

$\frac{4}{3} = e^{0.02t}$

$\ln\left(\frac{4}{3}\right) = 0.02t$ ①

$t = \frac{\ln\left(\frac{4}{3}\right)}{0.02}$

$t = 14.38 \dots$ ①

$\therefore$ In 15th year.

c) (i) ① $\uparrow$
Area = $\frac{5}{3} [1.2 + 2.2 + 4(2.5 + 4.2) + 2(3.1)]$ ①

$= 60\frac{2}{3} \text{ m}^2$ ①

(ii) Volume = $60\frac{2}{3} \times 6$

$= 364 \text{ m}^3$ ①

Question 14

a) (i) $800 \times 1.04^{14} = 1385.34$ ①

(ii) Total Investments =

$800 \times 1.04^{14} + 800 \times 1.04^{13} + \dots$

$800 \times 1.04^1 + 800$ ①

$S_n = \frac{800(1.04^{15} - 1)}{1.04 - 1}$ ①

$= \$16018.87$

(ii) $8000 \left(1 + \frac{r}{100}\right)^{14} = 16018.87$ ①

$1 + \frac{r}{100} = \sqrt[14]{\frac{16018.87}{8000}}$

$1 + \frac{r}{100} = 1.0508 \dots$

$\therefore r = 5.08\%$ ①

b) (i) $y = \frac{1 + \sqrt{x}}{2}$

$\sqrt{x} = 2y - 1$ ①

$\therefore x = (2y - 1)^2$

(ii) Volume =

$\pi \int_{\frac{1}{2}}^3 x^2 dy$

$= \pi \int_{\frac{1}{2}}^3 (2y - 1)^4 dy$ ①

$= \pi \left[\frac{(2y - 1)^5}{10} \right]_{\frac{1}{2}}^3$ ①

$= \frac{\pi}{10} [5^5 - 0]$

$= \frac{3125\pi}{10} = \frac{625\pi}{2}$

c) (i) $y = x^2 - 6x + 8$
roots $0 = x^2 - 6x + 8$
 $0 = (x - 2)(x - 4)$ ①
 $\therefore x$ intercepts are 2 + 4.

Area = $\int_0^2 x^2 - 6x + 8 + \left| \int_2^4 x^2 - 6x + 8 \right|$ ①
 $= \left(\frac{x^3}{3} - 3x^2 + 8x \right)_0^2 + \left(\frac{x^3}{3} - 3x^2 + 8x \right)_2^4$
 $= \left(\frac{8}{3} - 12 + 16 \right) - (0)$

$+ \left(\frac{64}{3} - 48 + 32 \right) - \left(\frac{8}{3} - 12 + 16 \right)$

$= \frac{20}{3} + \left| \frac{16}{3} - \frac{20}{3} \right|$

① $= \frac{20}{3} + \left| -\frac{4}{3} \right|$
 $= 8 \text{ units}$ ①

1 Mark for being aware of
splitting area into 2 parts.

1 mark for $\int_0^4 x^2 - 6x + 8$
 $= \frac{16}{3}$

Question 15

a) (i) $\ddot{x} = 4 \sin 2t$

$$\dot{x} = \int 4 \sin 2t = -\frac{4 \cos 2t}{2} + c$$

$$\dot{x} = -2 \cos 2t + c \quad (1)$$

when $t=0$ $\dot{x} = -1$

$$-1 = -2 \cos(0) + c \quad (1)$$

$$\therefore c = 1$$

$$\therefore \underline{\dot{x} = 1 - 2 \cos 2t}$$

(ii) At rest when $\dot{x} = 0$

$$1 - 2 \cos 2t = 0$$

$$\cos 2t = \frac{1}{2} \quad (1)$$

$$2t = \frac{\pi}{3}, \frac{5\pi}{3}, \dots$$

$$t = \frac{\pi}{6}, \frac{5\pi}{6}, \dots \quad (1)$$

$$\therefore \text{first at rest when } t = \frac{\pi}{6}$$

(iii) $x = \int 1 - 2 \cos 2t$

$$x = -\sin 2t + t + c_1$$

when $t=0$ $x=0$

$$0 = 0 + 0 + c_1 \quad (1)$$

$$c_1 = 0$$

$$\therefore \underline{x = t - \sin 2t}$$

(iv) (-0.34)

$$\begin{array}{ccc} \frac{\pi}{6} - \frac{\sqrt{3}}{2} & t=0 & \frac{\pi}{2} \\ \left(\frac{t-\pi}{6}\right)^2 & & t=\frac{\pi}{2} \\ \text{when } t = \frac{\pi}{6} & x = \frac{\pi}{6} - \frac{\sqrt{3}}{2} & (1) \end{array}$$

$$\begin{array}{l} \text{when } t = \frac{\pi}{2} \quad x = \frac{\pi}{2} - \sin 0 \\ \quad \quad \quad = \frac{\pi}{2} \quad (1.57 \dots) \end{array}$$

$\therefore$ Distance travelled

$$\begin{aligned} & 2 \left| \frac{\pi}{6} - \frac{\sqrt{3}}{2} \right| + \frac{\pi}{2} \quad (1) \\ & = 2 \left| -\left(\frac{\sqrt{3}}{2} - \frac{\pi}{6}\right) \right| + \frac{\pi}{2} \\ & = \sqrt{3} - \frac{\pi}{3} + \frac{\pi}{2} \\ & = \sqrt{3} + \frac{\pi}{6} \quad (2.255 \dots) \quad (1) \end{aligned}$$

If answer $\frac{\pi}{2} \rightarrow (1)$

If $\int_0^{\frac{\pi}{2}} \dot{x} - \sin 2t \rightarrow (1)$

1 of each error

b) (i) $e^{1-\ln 2} = e \cdot e^{-\ln 2}$
 $= e \cdot e^{\ln \frac{1}{2}} \quad (1)$
 $= e \times \frac{1}{2}$
 $= \frac{e}{2}$

(ii) $y' = -4e^{1-4x} \quad (1)$
 when $x = \frac{\ln 2}{4}$ $y' = -4e^{1-\ln 2} \quad (1)$
 $= -\frac{4e}{2} = -2e$

$$\therefore \int_1^e y - \frac{e}{2} = -2e \left(x - \frac{e}{2}\right) \quad (1)$$

$$4ex + 2y - e - 2e^2 = 0$$

c) Area = $\int_1^e (\ln 2x - \ln x) dx \quad (1)$
 $= \int_1^e \ln 2 + \ln x - \ln x \quad (1)$
 $= \int_1^e \ln 2$
 $= [x \ln 2]_1^e \quad (1)$
 $= e \ln 2 - \ln 2$
 $= \ln 2 (e - 1) \quad (1)$
 $(1.19 \dots)$

Question 16.

a) (i) $500 = x^2 y \therefore y = \frac{500}{x^2}$ ①

$$A = 4xy + x^2$$

$$\therefore A = 4x \left(\frac{500}{x^2} \right) + x^2$$
 ①

$$A = \frac{2000}{x} + x^2$$

(ii) Least Area when $A' = 0$
 $\& A'' > 0$

$$A' = -2000x^{-2} + 2x$$

$$2x - \frac{2000}{x^2} = 0$$

$$2x^3 = 2000$$

$$x^3 = 1000$$
 ①

$$x = 10$$

$$\text{Test } A'' = 4000x^{-3} + 2$$

$$= \frac{4000}{x^3} + 2$$

when $x = 10$ $A'' = 4 + 2$
 $= 6 > 0$ ①

$\therefore$ Least area occurs when $x = 10$

$$A = \frac{2000}{10} + 10^2$$

$$A = 300 \text{ a}^2$$
 ①

b (i) If A.P. ①

$$2\cos\theta - \sin\theta = 2\sin\theta - 2\cos\theta$$

$$4\cos\theta = 3\sin\theta \text{ --- (A)}$$

$$\therefore \tan\theta = \frac{4}{3}$$
 ①

$$\theta = 53^\circ 8'$$
 ①

(ii) from (A) $4\cos\theta = 3\sin\theta$
 $\therefore 2\cos\theta = \frac{3}{2}\sin\theta$ ①

$\therefore$ sequence becomes

$$\sin\theta, \frac{3}{2}\sin\theta, 2\sin\theta$$

hence next term is

$$\frac{5\sin\theta}{2}$$
 ①

c) (i) $m_{PA} = \frac{y-6}{x-1}$ ①

(ii) $m_{PB} = 2m_{PA}$

$$\therefore \frac{y-2}{x-3} = \frac{2(y-6)}{x-1}$$
 ①

$$(y-2)(x-1) = (x-3)(2y-12)$$

$$xy - y - 2x + 2 = 2xy - 12x - 6y + 36$$

$$xy - 5y - 10x + 34 = 0$$

$$y(x-5) = 10x - 34$$

$$y = \frac{10x - 34}{x - 5}$$
 ①

$$= \frac{10(x-5)}{x-5} + \frac{16}{x-5}$$

$$y = 10 + \frac{16}{x-5}$$
 ①

